Review Article

Global climate change and its impacts on the terrestrial Arctic carbon cycle with special regards to ecosystem components and the greenhouse-gas balance

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Abstract

The climatic changes on earth may have serious implications for the carbon (C) cycle in the terrestrial Arctic throughout the 21st century. Arctic vegetation takes up carbon dioxide (CO_2) from the atmosphere producing biomass. In a cold and often moist soil environment, dead organic matter is preferentially preserved as soil organic matter (SOM) due to the inhibition of decomposition processes. However, viable soil microbes exhale huge amounts of CO_2 and methane (CH_4) annually. Hence, Arctic ecosystems exhibit annual fluxes of both carbon-based (CO_2 and CH_4) greenhouse gases (GHGs) that are in an order of magnitude of millions of tons. Rising Arctic temperatures lead to the degradation of much of today's permafrost in the long run. As a result, large quantities of frozen SOM may become available for decomposers, and GHGs that are entrapped in permafrost may be released. At the same time, warming tends to stimulate the growth, development, and reproduction of many Arctic plants, at least transiently. The present northward migration of boreal shrubs and trees into southern tundra areas may be amplified by that, increasing the ecosystems' gross primary production and, thus, their C sequestration. On the other hand, rising temperatures boost SOM decomposition and microbial respiration rates. In general, soil temperature and soil moisture are key environmental variables to control the intensity of aerobic and anaerobic respiration by microbes, and autotrophic respiration by plants. On the basis of published data on Arctic CO_2 and CH_4 fluxes, the calculations on the terrestrial C-based Arctic GHG balance made in this review reveal a current annual GHG exchange that ranges between a weak storage of ≤ 225 Tg CO_2 equivalent (eq.) y^{-1} and a huge release of ≤ 1990 Tg CO_2 eq. y^{-1}. Hence, the Arctic GHG balance does apparently already contribute positively to the climatic changes at present. Regarding the future, the relative development of the uptake and release of CO_2 and CH_4 by northern ecosystems is fundamental to the overall GHG status of the Arctic under scenarios of continued climate change.

Key words: northern high latitudes / Arctic / climate change / CO_2 / CH_4 / carbon cycle / soil / greenhouse gas balance / review

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1 Introduction

The terrestrial northern high latitudes above 60° N are deeply connected with the global climate system (*ACIA*, 2005; *AMAP*, 2009), which is currently changing (*IPCC*, 2007a). In contrast to expectations, climatic changes in the Arctic are even more pronounced and extensive than previously thought (*AMAP*, 2009). A very large part of that can be attributed to the anthropogenic greenhouse-gas (GHGs) emissions of carbon dioxide (CO_2) and methane (CH_4) from fossil-fuel burning, agriculture, and land-use change, as both gas species are radiatively active in the atmosphere (*IPCC* 2001a, 2007a). Northern ecosystems are very sensitive to changes in climate due to serious ecological impacts of rising temperatures on Arctic vegetation (*e.g., McGuire* et al., 2002; *ACIA,* 2005; *Rennermalm* et al., 2005; *Kimball* et al., 2006) and soil microbial communities (*e.g., Vorobyova* et al., 1997; *ACIA*, 2005; *Schimel* and *Mikan*, 2005). Both ecosystem components are adapted to cold local conditions, and for the permafrost underlying vast areas in the Arctic, temperature is a key determinant, too (*e.g., Anisimov* and *Nelson*, 1997; *ACIA*, 2005; *Anisimov* and *Reneva*, 2006; *Romanovsky*, 2008). Other critical environmental variables are soil moisture (*e.g., Callaghan* et al., 2004a; *ACIA*, 2005; *Sitch* et al., 2007), nutrient availability (*e.g., Hobbie* et al., 2002; *Loya* et al., 2002; *Mack* et al., 2004; *ACIA*, 2005), and atmospheric CO_2 concentration (*e.g., Schimel*, 1995; *ACIA*, 2005; *Girardin* et al., 2008). Arctic vegetation and soil microbes are impor-

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tant for the carbon (C) cycle as they interchange huge amounts of CO_2 and/or CH_4 (*e.g., White* et al., 2000; *Callaghan* et al., 2004a; *Zhuang* et al., 2004, 2006; *ACIA*, 2005). High-latitude permafrost has an effect on both *via* active-layer depth and soil drainage. At present, 12.8%–17.8% of the Northern Hemisphere's land mass is characterized by permafrost (*Zhang* et al., 1999). Large sources of organic matter (OM) inside permafrost bodies may become accessible to decomposing organisms if permafrost degrades due to warming (*e.g., Davidson* and *Janssens*, 2006; *Field* et al., 2007; *Uhlířová* et al., 2007). In general, Arctic microbes are likely to increase the OM turnover and C release to the atmosphere under warmer conditions (*e.g., Hobbie* et al., 2002; *Mikan* et al., 2002; *Robinson*, 2002; *Marchand* et al., 2004; *ACIA*, 2005). On the other hand, plant species are predicted to increase their primary production if photosynthesis is becoming more intense (*e.g., McGuire* et al., 2002; *ACIA* 2005; *Rennermalm* et al., 2005; *Kimball* et al., 2006; *Stieglitz* et al., 2006). Consequently, climate change may have extensive effects on the Arctic C cycle including the total GHG balance based on CO_2 and CH_4.

In this article, we review the impacts of global climate change on the C cycle of the terrestrial Arctic by highlighting several ecosystem components as well as the fluxes of CO_2 and CH_4. After a brief statement on the present and projected future climatic changes in the Arctic (section 2), we regard the specific situation of northern ecosystems in a changing environment including the permafrost occurring there (sections 3 and 4). In section 5.1, various components and processes of the Arctic C cycle are discussed in detail. Published data on the annual range of CO_2 and CH_4 exchange are calculated in section 5.2 in order to give a quantification of the present status of the Arctic in terms of its total annual GHG balance (based on both gas species). Finally, the focus is laid on the ecosystems' long-term response to climate change concerning the ecological development and the C reservoirs that may be affected (section 6.1). The synthesis of information on future GHG fluxes from several publications presented here allows us to assess the current scientific knowledge on the potential future GHG exchange in the Arctic and its impacts on the climate system (section 6.2).

2 Present and future climatic changes in the Arctic

In the past 100 y, the regional warming rate in the Arctic was almost twice the global average (*IPCC*, 2007a). Between 1900 and 2003, the statistical warming of the local climate (≈ 0.09°C per decade; *ACIA,* 2005) was stronger than on the entire northern hemisphere (≈ 0.06°C per decade; *IPCC*, 2001a). During the last four decades, surface air temperatures increased by ≈ 0.04°C y^{-1} on average across the Arctic, while lower-latitude warming was 0.025°C y^{-1} (*ACIA*, 2004, 2005). Since the 1950s, the annual mean surface temperature in wintertime/springtime rose by ≈ 3°C–4°C in E Siberia and NE Canada / W Greenland (*ACIA* 2004, 2005). At present, Arctic summer temperatures are the highest of the last 400 y (*Chapin* et al., 2005).

For the future, the global mean temperature rise by 2090–2099, relative to 1980–1999, is projected to amount to 2.4°C on average (total range: 1.4°C–3.8°C) of the IPCC's B2 emission scenario; the rise in temperature concerning the A2 scenario is 3.4°C on average (total range: 2.0°C–5.4°C; *IPCC*, 2007a, b). Several general circulation models (GCMs) designated by *ACIA* (2005) simulate a global mean temperature rise by 2071–2090 in relation to 1981–2000 of ≈ 1.9°C on average (total range: 1.4°C–2.1°C); the corresponding warming of the Arctic accounts for 3.7°C on average (total range: 2.8°C–4.6°C; see Fig. 1). Winter temperatures are projected to increase more significantly than on the annual mean, ≈ 4°C–7°C across the terrestrial Arctic area (*ACIA*, 2004).

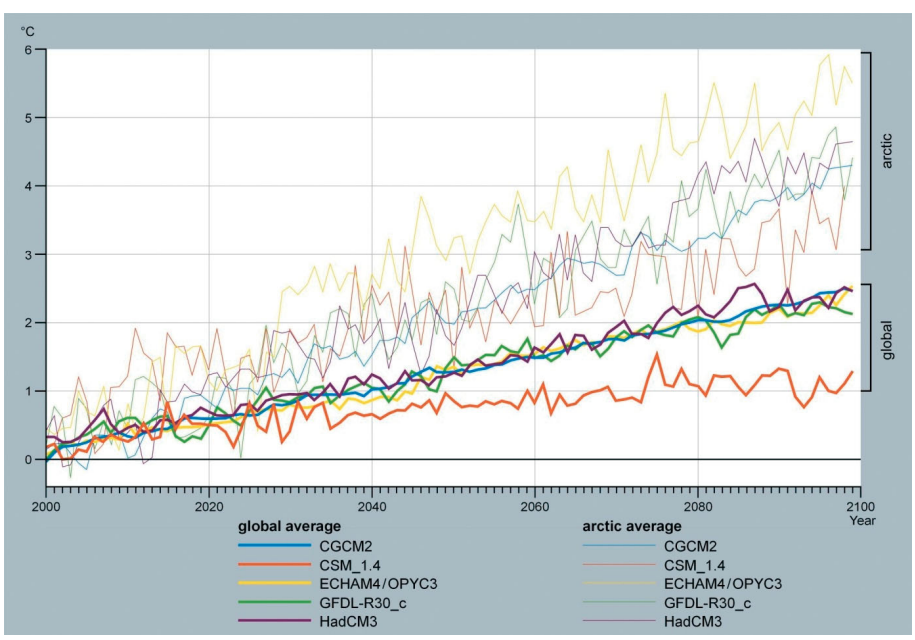

Figure 1: Changes in global and Arctic (60°–90° N) annual mean surface air temperature in relation to the baseline period 1981–2000 as projected by the five general circulation models (GCMs) designated by *ACIA* (2004, 2005) forced with the B2 emissions scenario (courtesy ACIA).

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An increasing transport of moisture to the northern high latitudes due to climate warming led to growing precipitation amounts (1.0%–1.4% on average per decade) between 1900 and 2003 (*ACIA*, 2005). Projections on precipitation development by 2071–2090 indicate an increase of 1.4%–4.7% (2.5% on average) for the global system and of 7.5%–18.1% (in the main: 10%–12%) for the Arctic area, which is predicted to concentrate particularly on the cold season, falling as rain predominantely (*ACIA*, 2004, 2005).

In the past, rising temperatures also led to a declining snow-cover extent throughout the Arctic (1%–2% y^{-1}). Snow-cover extent was on a 38-years low in 2007 as only 24.0 million km^2 were covered by snow (*AMAP*, 2009). In wintertime, increasing rainfalls on top of existing snow cover induce earlier snowmelt, and flooding situations if the rainfalls are intense (*ACIA*, 2004, 2005). Warming generally tends to cause earlier snowmelt in spring (*IPCC*, 2007a).

3 Response of permafrost to climate change

Permafrost is a characteristic feature of the cold northern high latitudes and defined as frozen soil, sediment, or rock having temperatures at or < 0°C for at least two consecutive years (*U.S. Arctic Research Commission Permafrost Task Force,* 2003, p. 3; *ACIA*, 2005). In summer, thawing at its surface generates a seasonal active layer, which is only a few decimeters deep in the High Arctic to more than 200 dm at southern margins of permafrost occurrance (*Steven* et al., 2006). The current distribution of permafrost originates from its historical development during the Pleistocene, its geographical position in terms of latitude, altitude, and distance from major water bodies, and the local environmental conditions, *e.g.,* surface-cover type (*Zhang* et al., 1999). Permafrost under exposed land surfaces, excluding glaciers, ice sheets, and water bodies, is divided into the four zones of continuous, discontinuous, sporadic, and isolated permafrost (Fig. 2). The total extent of all permafrost zones on the Northern Hemisphere is ≈ 22.79 × 10^6 km^2 or 23.9% of the hemispherical land mass (*Zhang* et al., 1999). Excluding all permafrost-free (unfrozen) areas, the real extent of permafrost amounts to 12.21 × 10^6 to 16.98 × 10^6 km^2 or 12.8%–17.8% of the land mass (*Zhang* et al., 2000). The current thickness of subsurface permafrost ranges from a few decimeter at its southern parts to 740 m in Alaska and 1500 m in Siberia (*U.S. Arctic Research Commission Permafrost Task Force,* 2003, p. 6; *Anisimov* and *Reneva*, 2006).

As part of the cryosphere, permafrost develops and exists only in cold conditions below or near the freezing point. Much of today's permafrost is in thermal disequilibrium concerning current climate conditions owing to the warming of the Arctic (*Anisimov* and *Nelson*, 1997). Permafrost temperatures rose in several areas during the past decades (*Richter-Menge* et al., 2006; *AMAP*, 2009). In Alaska for example, permafrost warmed by ≈ 1°C–3°C above a 70-years average (1930–2003). At some sites temperatures are close to 0°C meanwhile (*Romanovsky*, 2008). Where soil temperatures already passed the 0°C mark, thawing of the permafrost at its margins has started, accompanied by a deepening of the active layer (*ACIA* 2004, 2005; *Anisimov* and *Reneva*, 2006; *Romanovsky*, 2008; *AMAP*, 2009). However, data on active-layer-depth development do not suggest a consistent trend across the Arctic region (*Richter-Menge* et al., 2006). External influences such as wildfires, floods, or anthropogenic activities, *e.g.,* agriculture and infrastructure, triggered permafrost thawing in much more northern locations than caused by the internal increase in ground temperatures (*Richter-Menge* et al., 2006). Nevertheless, if climate warming continues in the future, large permafrost areas are likely to thaw and degrade in the long run. Simulations based on the A1B emission scenario (global mean temperature rise of 1.7°C–4.4°C from 1980–1999 to 2090–2099, *IPCC*, 2007b) project an increase in soil temperatures of 4.1°C in the Russian Arctic to 6.6°C in the Canadian Arctic between the 1990s and 2090s (Community Climate System Model, CCSM3; *Kitabata* et al., 2006). Such an intensive warming could lead to a degradation of the current permafrost of ≈ 15% to a maximum of 50% until 2050 (*Anisimov* and *Reneva*, 2006). Corresponding increases in active-layer depth by the end of this century are likely to reach 20%–50% or even more in the northernmost locations (*Anisimov* et al., 2002; *Stendel* and *Christensen*, 2002; *ACIA*, 2005; *Anisimov* and *Reneva*, 2006). On the continental scale, an irrecoverable retreat of the residual permafrost of ≈ 150–200 km to the north may occur (*Anisimov* et al., 2002; *ACIA*, 2004, 2005). Finally, this could have far-reaching consequences for those ecosystems that are affected by permafrost degradation.

4 Cold northern ecosystems and their conditions for life

Northern high-latitude ecosystems have long been adapted to the extreme climatic demands of this geographical region. The 10°C July-isotherm at 70° N is a striking circumpolar climatic-ecological border between the tundra ecosystems in the north and the boreal forests (taiga) further south (*Schultz*, 2000). The present geographical distribution of both biomes on the northern hemisphere is shown in Fig. 2. In the north, tundra ecosystems cover some 5.0 × 10^6 to 10.5 × 10^6 km^2 and differ ecologically among each other with regard to their specific floristic societies of algae, lichens, mosses, grasses (herbs), sedges, or dwarf-shrubs (*Joabsson* and *Christensen*, 2001; *ACIA*, 2005). Their flat-growing vegetation changes to dwarf-shrubs and small trees towards south, at last giving way to the vast boreal forests (*Callaghan* et al., 2004a; *ACIA*, 2005). Ecologically, the taiga forms the largest part of the northern high latitudes (and reaches further south than 60° N), covering 14.3 × 10^6 to 20.0 × 10^6 km^2 (*Turetsky*, 2004; *Wieder* et al., 2006, p. 4). In lowland areas, many tundra and borel ecosystems exhibit anoxic conditions owing to lasting water logging. These areas are often wetlands such as swamps or peatlands. Some 3 × 10^6 to 4 × 10^6 km^2 of northern peatlands are located within boreal lowlands (*Vitt*, 2006, p. 9; *Wieder* et al., 2006, p. 5; *Frolking* and *Roulet*, 2007).

Most tundra ecosystems, and many boreal ones, are permafrost environments, which are ecologically characterized by a spezialized group of cold-adapted microbes that are viable at temperatures below the freezing point (*Vorobyova* et al., 1997; *Steven* et al., 2006). These microbial communities

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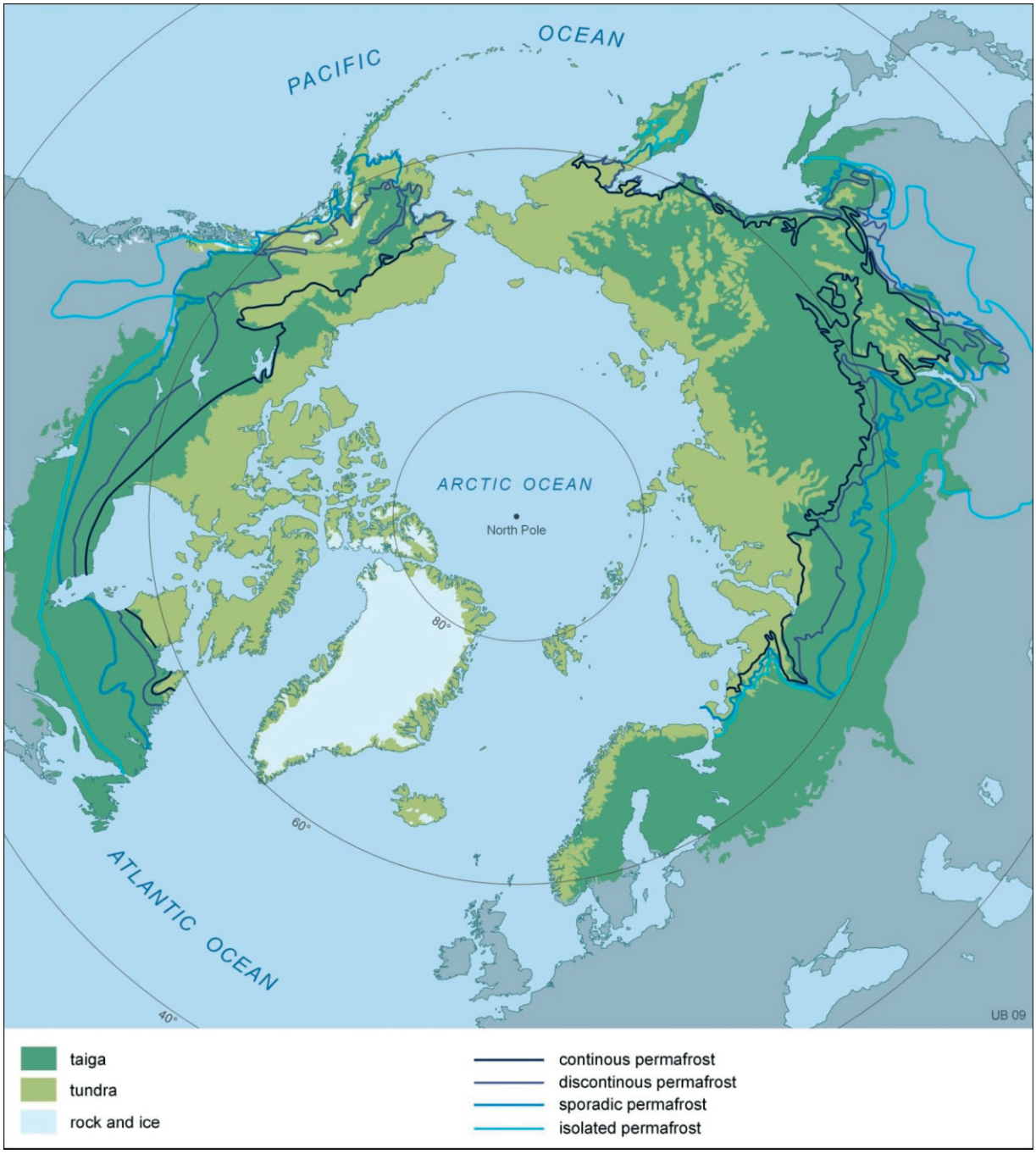

Figure 2: Current geographical extent of the tundra ecosystems and the taiga on the Northern Hemisphere (modified from *Olson* et al., 2001). The thin lines mark the southern borders of the continuous, discontinuous, sporadic, and isolated permafrost zone (modified from *Brown* et al., 1998, revised Feb. 2001).

resisted the demanding life conditions in permafrost for up to several millions of years (*Vorobyova* et al., 1997). A phase of dormancy (*Schimel* and *Mikan*, 2005) allowed them to reduce their metabolic cell activity to a minimum during the long-term impact of the cold (*Vorobyova* et al., 1997; *Steven* et al., 2006). However, some species are able to metabolize at temperatures down to –39°C (*Panikov* et al., 2006). The microbial diversity in Arctic habitats is generally considerable: Polar desert and tundra soils contain as much species as taiga soils further south, exceeding arable soils by far; compared to other major biomes, many species of bacteria, fungi, and algae are less in numbers or even absent, *e.g.,* actinomycetes (*ACIA*, 2005).

In the context of the terrestrial Arctic C cycle, all C-respiring microbes are important: These are particularly methanogenic *archaea* as CH_4 producers, methanotrophic *bacteria* as CH_4 consumers and CO_2 producers (both *cf.* section 5.1.3) as well as all other CO_2-respiring species (section 5.1.2). As long as these microbes maintain their dormant state within

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the cold permafrost habitats, microbial C consumption and respiration is low. Warming and thawing of the permafrost is very likely to be favorable to the metabolic activity, growth, and reproduction of most microbial communities (*Vorobyova* et al., 1997; *ACIA*, 2005; *Schimel* and *Mikan*, 2005). A return to their full potential to decompose and metabolize OM under warming conditions (*Vorobyova* et al., 1997; *ACIA*, 2005) may lead to significant implications for the C turnover and other important biogeochemical processes in northern ecosystems (*Lawrence* and *Slater*, 2005; *Steven* et al., 2007).

5 The present carbon cycle in the terrestrial northern high latitudes

About 50% of the C that has been photosynthetically assimilated by plants or taken up by microbes is finally integrated into the organisms' tissue as part of their biomass (*Luo*, 2007). Among the most important factors for photosynthesis are photosynthetically active radiation (*Davidson* and *Janssens*, 2006; *Groendahl* et al., 2007), temperature (*Illeris* et al., 2004; *Groendahl* et al., 2007; *Luo*, 2007), soil moisture (*Davidson* and *Janssens*, 2006), nutrient availability (*Mikan* et al., 2002; *Marchand* et al., 2004), atmospheric CO_2 concentration, and growing-season length (*Luo*, 2007; *Girardin* et al., 2008). The intensity of the photosynthesis and, hence, the amount of C being assimilated by plants is currently changing due to the fact that these variables are all affected by the current climatic changes. Beside the C uptake *via* photosynthesis (gross primary production, GPP), plants release CO_2 to the atmosphere (autotrophic respiration). On the ecosystem scale, the ratio of vegetation C assimilation to vegetation C release is important, yielding an ecosystem's net primary production (NPP). The annual NPP of the tundra and boreal vegetation as well as in biomes further south are summarized in Tab. 1.

Table 1: Annual net primary production (NPP) of the vegetation in the tundra and taiga in comparison to ecosystems in biomes further south (according to *Schultz*, 2000).

Kind of ecosystems	Annual NPP
Ecosystems in the tundra	< 1–500 g C m^{-2} y^{-1}
Ecosystems in the taiga	400–800 g C m^{-2} y^{-1}
Ecosystems in biomes further south	< 2900 g C m^{-2} y^{-1}

The average pool of C being sequestered by the standing vegetation of different Arctic ecosystems is given in Tab. 2.

Table 2: Average pool of C sequestration by the standing vegetation of different northern ecosystems compared with the global terrestrial biosphere (according to *Callaghan* et al., 2004b; *ACIA*, 2005).

Kind of ecosystem	Average amount of C being sequestered
Arctic / Alpine ecosystems	550 g C m^{-2}
Boreal woodlands	4150 g C m^{-2}
Boreal forests	9450 g C m^{-2}
Global terrestrial biosphere	7150 g C m^{-2}

In addition to autotrophic respiration, many microbes exhale CO_2 or CH_4, which is lost to the atmosphere. Hence, heterotrophic respiration reduces the amount of net C sequestration by land ecosystems, yielding the net ecosystem productivity (NEP; *Schulze* et al., 2000; *ACIA*, 2005; *Lovett* et al., 2006; *Luo*, 2007). High-latitude ecosystems such as wetlands (peatlands) accumulated large amounts of organic C in the past owing to a positive NEP as ecosystem C uptake exceeded ecosystem C release.

5.1 The most important components and processes of the terrestrial Arctic carbon cycle

5.1.1 Soil organic C and dissolved organic C

Dead plants and animals as well as microbial materials and their decomposition products of different size, shape, and biochemical composition, from fresh and unaltered materials to OM that has been significantly altered by microbial decomposition, are basic products for the formation of soil organic matter (SOM) in northern ecosystems (*Sollins* et al., 1996; *Baldock*, 2007, p. 2). The process of microbial decomposition leads to the mineralization of SOM into inorganic ones. This process includes the transformation of soil organic carbon (SOC), which forms about half of the SOM, into GHGs such as CO_2 or CH_4, and other products. Alternatively, SOM can be altered by humification processes, which decrease the decomposability of the affected OMs, favoring their stabilization as SOM (*Sollins* et al., 1996; *Robinson*, 2002; *Baldock*, 2007, p. 17). The biochemical quality of SOM is an important parameter for its decomposability as decomposers break down labile components preferentially, *e.g.,* carbohydrates, thus selectively preserving components that decompose much more slowly and becoming recalcitrant (*Hobbie* et al., 2000; *Kögel-Knabner*, 2002). Beside SOM quality, several environmental variables such as soil temperature (*Hobbie* et al., 2000, 2002; *White* et al., 2004), soil moisture (*Hobbie* et al., 2000, 2002; *Robinson*, 2002; *White* et al., 2004), soil pH, and nutrient availability (*White* et al., 2004), and the activity of decomposing organisms (*Robinson*, 2002) have an effect on the SOM alteration. Microbial decomposition is particularly sensitive in cold regions since the relative sensitivity of this process to increasing temperatures is greater at low soil temperatures (*Kirschbaum*, 1995). Low temperatures as well as an extremely high or low soil moisture inhibit decomposition processes in Arctic soils, leading to the long-term preservation of SOM (*Hobbie* et al., 2000, 2002; *Mikan* et. al., 2002; *Robinson*, 2002; *AMAP*, 2009). According to *McGuire* et al. (2006), cold and/or anoxic soil environments are responsible for the predominant protection of SOM from decomposition in a lot of Arctic ecosystems. The slow decomposability of the OMs *per se* is only of minor importance in this context.

Topography and soil material are also critical because mineral upland soils generally favor decomposition due to good soil drainage and aeration. These soils often contain less SOM and have lower C stores compared to moist or wet lowland soils. Decompostion rates in wetlands are diminished as a result of frequent anoxia caused by water logging, which induces the accumulation of deep SOM layers with high C contents (*Hobbie* et al., 2000; *Davidson* and *Janssens*, 2006).

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Soil organic C forms the largest reservoir of the terrestrial Arctic C cycle. Northern soils store huge amounts of it owing to the unfavorable soil environments as particularly in wetlands (*Friborg* et al., 2003; *ACIA*, 2005; *Dutta* et al., 2006; *Rodionov* et al., 2007). Estimates of the amount of SOC being stored in the northern high latitudes are presented in Tab. 3.

Table 3: Estimated stocks of soil organic carbon (SOC) in petagram (Pg) in the northern high latitudes and on the global scale.

Kind of ecosystems	Range of estimated SOC stocks
Northern wetlands	120–460 Pg C[a]
Global wetlands	202–535 Pg C[b]
Tundra ecosystems	43–200 Pg C[c]
Boreal ecosystems	200–750 Pg C[d]
Northern high latitudes > 45° N	1400–1850 Pg C[e]

[a] *Gorham* (1991); *Hobbie* et al. (2000); *Turunen* et al. (2002)
[b] several authors in *Batjes* (1996); *Mitra* et al. (2005)
[c] *Hobbie* et al. (2000); *Stieglitz* et al. (2006)
[d] *Hobbie* et al. (2000); *Rodionow* et al. (2006); *Stieglitz* et al. (2006)
[e] *AMAP* (2009)

During wildfires, smaller proportions of the SOM are converted to charred black carbon (*e.g., Czimczik* et al., 2003), which is considered to be a recalcitrant part of SOM. *Ohlsen* et al. (2009) estimate that boreal forest soils store 1 Pg C as black carbon being equivalent to 1% of the total vegetation C stocks in this biome.

Cryoturbation is a key soil process to incorporate large amounts of SOM into permafrost soils potentially, particularly into mineral ones, resulting in a long-term storage of SOM as a consequence of impeded decomposition rates and inaccessability to most microbes (*Hobbie* et al., 2000, *Rodionow* et al., 2006; *Kaiser* et al., 2007; *Tarnocai* et al., 2007, p. 130). The preserving effect of cryoturbation is much stronger if the SOM is entirely encased in the permafrost (*Kaiser* et al., 2007; *Tarnocai* et al., 2007, p. 130). Soil organic-C stocks of permafrost soils are estimated to 350–450 Pg C (*Stokstad*, 2004; *Davidson* and *Janssens*, 2006). Frozen loess deposits (yedoma) have been assessed to 450–500 Pg C (*Zimov* et al., 2006a, b), while steppe-tundra soils may contain some 720 Pg C and sediments beneath ice sheets some 380 Pg C (*Zimov* et al., 2006a, 2006b). Due to its preservation, certain proportions of the organically bound C belong to the pool of potentially mineralizable SOC (*Neff* and *Hooper*, 2002; *Weintraub* and *Schimel*, 2003; *Uhlířová* et al., 2007). Expecting climate warming and permafrost thawing to continue in the future, the mineralizable SOC may become decomposed, increasing respiratory C losses from Arctic ecosystems (*AMAP*, 2009).

Dissolved organic matter (DOM) is a complex mixture of organic compounds ranging from low-molecular acids to high-molecular polyelectrolytes (*Kaiser* et al., 2002). Sources for DOM are root exudates, microbial biomass, plant litter, or humus (*e.g., Hagedorn* et al., 2004). Formation, properties, and movement of DOM through the soil profile are influenced by permafrost and active-layer depth (*Kalbitz* et al., 2000; *Blodau*, 2002; *Kawahigashi* et al., 2004, 2006; *Frey* and *Smith*, 2005; *Prokushkin* and *Guggenberger*, 2007). In mineral soils, concentrations and fluxes of DOM in soil solution often decrease significantly with soil depth as the soil mineral phase (clay minerals, Fe and Al oxides/hydroxides) tends to adsorb the recalcitrant, hydrophobic compounds in favor (fractionation). This retentive effect through sorption enriches the hydrophobic fraction in mineral subsoils. The labile, hydrophilic fraction becomes microbially mineralized and/or exported from soils *via* leaching (*Kalbitz* et al., 2000; *Guggenberger* and *Kaiser*, 2003; *Kawahigashi* et al., 2004, 2006). Permafrost regime has a decisive influence on the DOM dynamics in soil solution: Permafrost defines the magnitude of DOM sorption as the active-layer depth is fundamental to mineral-layer thickness where sorption occurs. In general, sorption is most intensive in southern parts of permafrost distribution as well as in summer months, where/when the active layer has its maximum depth and, thus, allows for the largest sorption capacity in mineral subsoils (*Kawahigashi* et al., 2004, 2006; *Prokushkin* and *Guggenberger*, 2007). Like SOC, dissolved organic carbon (DOC), about half of the DOM, is another large reservoir of the C cycle. Each year, ≈ 10% of the DOC entering the world oceans *via* land runoff is discharged into the Arctic Ocean by river systems (*AMAP*, 2009). Huge amounts of that originate from northern peatlands (*AMAP*, 2009) where DOC concentrations are 20–60 mg L^{-1} and export rates to Arctic rivers amount to 1–50 g m^{-2} y^{-1} (*Blodau*, 2002).

5.1.2 Carbon dioxide

The composition of plant species, *i.e.,* the vegetation type, is important for the total GPP and NPP of an ecosystem, and has an effect on microbial respiration (mineralization) *via* SOM quality (*Grogan* and *Chapin*, 1999; *Ström* and *Christensen*, 2007). Warming tends to enhance primary production, even though there is a downregulation of the photosynthetic process at higher temperatures (*Field* et al., 2007; *Luo*, 2007). In general, ecosystem CO_2 release is controlled by several environmental variables such as soil temperature or soil moisture, which act together to determine respiration rates (*Robinson*, 2002). Plant and microbial respiration respond in different ways to temperature fluctuations (*Mikan* et al., 2002). The Q_{10} values of the aerobic mineralization (CO_2 production) vary over a wide range from 0.7 to 237.0 at different soil temperatures and high-latitude locations (*Hobbie* et al., 2000; *Blodau*, 2002; *Mikan* et al., 2002; *Bölter* et al., 2003, p. 18). Aerobic respiration activity generally benefits from rising temperatures (*Mikan* et al., 2002; *Illeris* et al., 2004). Soil moisture exerts an important influence on microbial CO_2 release as the aerobic respiration does only occur in the oxic soil zone. Reducing soil moisture from saturation to some 60% can lead to an increase in respiration rates. If soil moisture declines further, respiration rates decrease (*Robinson*, 2002; *Marchand* et al., 2004). In wetlands (peatlands), water-table position is important for CO_2 fluxes: If they fall below the soil surface, soil aeration and intensify CO_2 efflux improve (*Robinson*, 2002). Rising to or above the soil surface allows anaerobic processes to prevail (section 5.1.3).

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All influences on autotrophic and heterotrophic respiration are key to the spatial and temporal variability of CO_2 fluxes between northern ecosystems and the atmosphere. On an annual time scale, CO_2 uptake in Arctic areas is limited to the vegetation-growing season (*Blodau*, 2002; *Rennermalm* et al., 2005). In summertime, many ecosystems tend to be seasonal sinks due to a positive NEP (*Rennermalm* et al., 2005). During that time of the year, some 20%–93% of the CO_2 efflux originates from plant roots (*Bölter* et al., 2003, p. 16; *Raich* et al., 2002). High amounts of precipitation normally amplify photosynthesis and generate strong sinks. On the other hand, dry, hot summers induce oxic soil conditions, which increase aerobic respiration, while primary production is restrained by high temperatures and water shortage. Then respiratory losses outrange photosynthesis, shifting the positive CO_2 summertime exchange towards a weak sink, or sometimes even towards a source (*Blodau*, 2002; *Rennermalm* et al., 2005). As heterotrophic CO_2 release can occur throughout the year (*Rennermalm* et al., 2005), northern ecosystems can be substantial sources (negative NEP) in wintertime (*Mikan* et al., 2002; *Marchand* et al., 2004; *Panikov* et al., 2006). As a result, 3%–65% of the total annual CO_2 efflux originate from cold-season respiration (*Hobbie* et al., 2000; *Kurganova* and *Tipe*, 2003).

The long-term inhibition of SOM mineralization and CO_2 production generated CO_2 sinks in tundra and boreal wetlands (peatlands) in the past (*Blodau* and *Moore*, 2003; *Ström* et al., 2003; *Ström* and *Christensen*, 2007; *AMAP*, 2009). Their annual CO_2 uptake is estimated to 200 Tg (1 Tg = 10^{12} g) CO_2 y^{-1} (*Frolking* and *Roulet*, 2007). Regionally, tundra ecosystems have been both sinks and sources due to a highly variable CO_2 exchange over time and space (*Sitch* et al., 2007). For the entire 20th century, net CO_2 exchange between tundra ecosystems and the atmosphere was slightly positive, resulting in a net storage (*Sitch* et al., 2007). In retrospective simulations with the global vegetation model "hybrid v4.1", an annual CO_2 sink has been found for the high latitudes north of 50° N, which has a present average of ≈ 400 Tg CO_2 y^{-1} per decade (*White* et al., 2000). Continuous increases in the amount of CO_2 being stored since the starting date 1860, appearing especially after 1960, have been simulated (*White* et al., 2000). In *AMAP* (2009), the annual amount taken up by plants is estimated to 300–600 Tg CO_2 y^{-1} for the land area above 45° N, which is ≈ 30%–60% of the global sink capacity for atmospheric CO_2. For the time period 1921–1994, *McGuire* et al. (2000) simulate a fluctuating annual CO_2 exchange (NEP) above 50° N using the "Terrestrial Ecosystem Model" (TEM). Model outputs ranged from a loss of 18 g C m^{-2} y^{-1} to a gain of 16 g C m^{-2} y^{-1}, which indicates a long-term release of 1 g C m^{-2} y^{-1} until the last 25 y when the NEP is positive (10 g C m^{-2} y^{-1}). For the present, a CO_2 sink is attributed to the Arctic by *AMAP* (2009).

Northern ecosystems that were exposed to a weak drying or wetting in the past kept their CO_2-sink status or increased their net storage. Those ecosystems experiencing warming and drying acted as sources for at least some time (*Callaghan* et al., 2004a; *AMAP*, 2009). In the Alaskan tundra for example, ecosystems began to loose substantial amounts of CO_2 in the early 1980s when the local climate began to warm and dry (*Oechel* et al., 2000). After the period 1983–1990 when CO_2-exchange measurements revealed a loss in summer (*Oechel* et al., 1993), ecosystems began to sequester C slightly in summertime once again. However, they remained sources on an annual time scale due to the much higher CO_2 loss in wintertime (*Oechel* et al., 2000). For the entire northern high latitudes above 50° N, TEM simulation results from *Zhuang* et al. (2006) indicate a small annual CO_2 loss to the atmosphere in the amount of 240 Tg C y^{-1} (213 Tg C y^{-1} from forest fires) at present climate conditions. The annual CO_2 efflux from the global pedosphere has been assessed to 50–80 Pg CO_2 y^{-1} (*Kudeyarov* and *Kurganova*, 2005; *Raich* et al., 2002).

5.1.3 Methane

In extremely reducing soil environments, a highly specialized community of anaerobic microbes breaks down the SOM. During the last stage of the anaerobic decomposition, methanogenic *archaea* (methanogens) reduce several precursors to CH_4 (methanogenesis). Methane that moves towards the atmosphere passes the oxic soil zone where it can be assimilated by methanothrophic *bacteria* (methanotrophs). These CH_4-consuming bacteria live at oxic–anoxic interfaces and act as a biological filter for CH_4: After its consumption, the CH_4 is oxidized to CO_2 internally. Although these microbes reduce the amount of the CH_4 emission, their activity is responsible for an enhancement of the CO_2 efflux at the same time (*Topp* and *Pattey*, 1997; *Trotsenko* and *Khmelenina*, 2005; *Whalen*, 2005).

The ratio of CH_4 production to CH_4 consumption and oxidation is of great importance for the CH_4 exchange as part of the total GHG balance. Transport is a key process of the CH_4 dynamics in northern ecosystems: Molecular gas diffusion through the dissolved soil phase or through wetland water bodies, and subsequent loss into the atmosphere, is a minor pathway for the CH_4 to escape from soils (*Topp* and *Pattey*, 1997; *Whalen*, 2005). The rise of CH_4-containing bubbles through water bodies (ebullition) is another process of CH_4 release. The episodic efflux is very heterogeneous in space and time and becomes activated by changing atmospheric temperatures, changing atmospheric pressure, and by falling or rising water tables (*Topp* and *Pattey*, 1997; *Whalen*, 2005). The bubbles may contain CH_4 in quantities, which exceed the mass dissolved in the liquid phase by a factor of three. Ebullition has been made accountable for < 1% to 17% of the total CH_4 emissions from different northern sites (*Whalen*, 2005). Vascular plants have the most important influence on CH_4 fluxes as they are conduits for internal gas diffusion *via* their aerenchymatous tissue. This plant-internal process allows the CH_4 to bypass the oxic soil zone (*Topp* and *Pattey*, 1997; *Ström* et al., 2003; *Whalen*, 2005). The strength of the plant-mediated diffusion depends on the plant species and their abundance, *e.g.,* the abundance of sedges. Overall, this process can account for up to 90% of the CH_4 emissions from northern ecosystems (*Topp* and *Pattey*, 1997; *Blodau*, 2002).

Temperature, soil moisture (water-table depth), and vegetation type, which are all affected by permafrost regime and active-layer depth, have an influence on the production,

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transport, and oxidation of CH_4 (*Friborg* et al., 2000; *Ström* and *Christensen*, 2007; *Wille* et al., 2008). Soil temperature is a key determinant of CH_4 production and oxidation rates as methanogens and methanotrophs are both sensitive to temperature variations. In general, methanogenesis has a much higher temperature response compared to methanotrophy (*Topp* and *Pattey*, 1997; *Blodau*, 2002; *Callaghan* et al., 2004b; *ACIA*, 2005; *Whalen*, 2005). The Q_{10} values of the anaerobic CH_4 production range from 1 to 35, while values for oxidation are between 1.1 and 2.9 (*Blodau*, 2002; *Whalen*, 2005). At rising wetland water tables, soil conditions for methanogenesis improve as long as the water tables remain below the soil surface. Conversely, rising water tables reduce methanotrophy, thus enhancing total CH_4 emissions from wetlands (*Whalen*, 2005; *Ström* and *Christensen*, 2007). Water tables that rise aboveground, and lead to the formation of water bodies, do no longer increase CH_4 production; the efflux becomes hampered as the transport through water by molecular gas diffusion or ebullition is less efficient compared to air transport (*Zona* and *Oechel*, 2008). Atmospheric parameters such as friction velocity and atmospheric pressure are also main predictors for daily variations of CH_4 emissions on the landscape scale, suggesting a strong dependence of the actual transport on interactions with the atmosphere (*Sachs* et al., 2008; *Wille* et al., 2008). Vascular transport of oxygen (O_2) from the atmosphere into the rhizosphere does not only suppress methanogenesis, but also facilitates methanotrophy simultaneously (*Topp* and *Pattey*, 1997; *Joabsson* and *Christensen*, 2001; *Ström* et al., 2003; *Whalen*, 2005). Oxidation can occur either within the rhizosphere or during transport to the atmosphere. Ultimately, 20%–50% of the CH_4 being produced in northern ecosystems is subject to oxidation (*Callaghan* et al., 2004b; *ACIA*, 2005; *Whalen*, 2005); according to *Liebner* and *Wagner* (2007), 43%–90%, of the CH_4 is actually converted to CO_2.

Most northern wetlands (peatlands) as well as many moist tundra ecosystems acted as long-term sources for atmospheric CH_4 in the past, which is a result of the prevailing anoxia and the abundance of vascular plants (*Joabsson* and *Christensen*, 2001; *Blodau* and *Moore*, 2003; *ACIA*, 2005; *Frolking* et al., 2006; *Ström* and *Christensen*, 2007). In general, CH_4-emission rates can be very divergent: An Alaskan wet meadow tundra and shrub/tussock tundra for example emitted CH_4 in summertime at rates of 100–700 mg CH_4 m^{-2} d^{-1} (*Callaghan* et al., 2004a; *ACIA*, 2005). At a site in high-Arctic NE Greenland, *Friborg* et al. (2000) measured midsummer emission rates from virtually no flux to more than 120 mg CH_4 m^{-2} d^{-1}. In a Siberian Arctic polygonal tundra, *Sachs* et al. (2008) and *Wille* et al. (2008) found midsummer efflux rates of ≈ 17.8 mg m^{-2} d^{-1} and 30 mg m^{-2} d^{-1} on average, respectively. Unusually high rates can attain > 1000 mg CH_4 m^{-2} d^{-1} and are restricted to wetland water bodies (*Blodau*, 2002). The highest CH_4 fluxes in the northern high latitudes have been observed at warmer anoxic soils of the taiga (*Callaghan* et al., 2004a). Although boreal wetlands show the highest efflux rates in simulations (23 g CH_4 m^{-2} y^{-1}), wet tundra ecosystems represent a larger overall source due to their larger overall area (*Zhuang* et al., 2004). Cold-season fluxes of CH_4 from northern wetlands account for 3.5%–40.0% of the annual CH_4 exchange (several authors in *Wille* et al., 2008). Contributions of fluxes to the overall ecosystem GHG balance have been estimated to be in the amount of 14%–19% (*Friborg* et al., 2003; *Wille* et al., 2008).

On an annual time scale, some 10–63 Tg CH_4 y^{-1} are emitted from northern wetlands (*Callaghan* et al., 2004a; *Anisimov* and *Reneva*, 2006; *Frolking* and *Roulet*, 2007); estimates for the total global wetland emission vary between 60 and 237 Tg CH_4 y^{-1} (*IPCC*, 2001b, 2007; *Gedney* et al., 2004; *Mitra* et al., 2005; *Whalen*, 2005; *Anisimov* and *Reneva*, 2006). Assessments on the annual efflux from the entire northern high latitudes total 15–70 Tg CH_4 y^{-1} (*Callaghan* et al., 2004a; *Zhuang* et al., 2004, 2006; *ACIA*, 2005; *Metje* and *Frenzel*, 2007; *AMAP*, 2009), while site study values of land and freshwater fluxes attain 31–100 Tg CH_4 y^{-1} (*AMAP*, 2009).

In dry upland ecosystems, as particularly in boreal upland soils, good drainage and soil aeration induce a net uptake of atmospheric CH_4 (*Topp* and *Pattey*, 1997). In a new version of the TEM, simulated amounts of net uptake in the Arctic total 6.3 Tg CH_4 y^{-1} during the 1990s, which is higher than in most other estimates for that region (0–5.5 Tg CH_4 y^{-1}, *Zhuang* et al., 2004). The global sink capacity for atmospheric CH_4 by soils has been projected to 10–44 Tg CH_4 y^{-1} (*IPCC*, 2001b; *Whalen*, 2005).

5.2 Assessing the range of the present greenhouse-gas emissions in the terrestrial Arctic

Relative contributions of CH_4 to the total GHG efflux (CO_2 and CH_4) from the Arctic range from < 1% in dry ecosystems to > 20% at some wet tundra sites; common contributions from wet tundra ecosystems and northern wetlands attain 2%–10% (*Callaghan* et al., 2004a; *ACIA*, 2005). On the global scale, CH_4 contributions are highest from northern wetlands (*Whiting* and *Chanton*, 2001). As CH_4 contributes much stronger to the greenhouse effect per unit mass than CO_2 (*IPCC*, 2001b, 2007a), its release into the atmosphere is crucial to the progress of future climate change. In order to determine the impact of both gas species on the earth's radiative forcing and thus on climate, the total CH_4 emission is converted into an emission of CO_2 equivalents (CO_2 eq.). For this purpose, the total CH_4 emission is multiplied with the global-warming potential of CH_4 (GWP_{CH4}) yielding the corresponding amount in CO_2 eq. (*Frolking* et al., 2006). The effect of CH_4 on the earth's radiative forcing which is expressed by the GWP depends on the time horizon being chosen: On a horizon of 20 y, the GWP_{CH4} has a value of 72. Regarding a horizon of 100 y the GWP_{CH4} is 25, and for the 500 y horizon it is 7.6 (*IPCC*, 2007a). According to that, the amount of CO_2 equivalent (eq.) that is comparable to an emission of 1 Tg CH_4 totals 72 Tg CO_2 eq. on a horizon of 20 y. Regarding 100 y, 1 Tg CH_4 is the same as 25 Tg CO_2 eq., and after 500 y, the corresponding amount of CO_2 eq. would be 7.6 Tg CO_2 eq. Hence, CH_4 is a short-lived climate forcer, whose influence on the radiative forcing is very strong on short time horizons. After 20 y, CO_2 starts to dominate the radiative forcing impact of both GHGs in the atmosphere (*Frolking* et al., 2006).

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For a more accurate assessment of the total annual GHG balance in the terrestrial northern high latitudes, published data on the annual CO_2 and CH_4 exchange have been synthesized. In this context, available data on the annual CO_2 flux in the Arctic (*cf.* section 5.1.2) range from a sink of 300–600 Tg CO_2 y^{-1} (case A in Tab. 4) to a source of 240 Tg CO_2 y^{-1} (case B in Tab. 4). Data on the annual CH_4 flux (*cf.* section 5.1.3) indicate a net loss of 15–70 Tg CH_4 y^{-1}, which is equivalent to a release of 375–1750 Tg in CO_2 eq. y^{-1} for a GWP_{CH4} of 25. For the total GHG balance, CO_2- and CH_4-flux data are combined: Thus, totalling the CO_2 sink and the CH_4 source results in a total annual GHG exchange that varies from a net uptake of 225 Tg CO_2 eq. y^{-1} to a net release of 1450 Tg CO_2 eq. y^{-1} (result A in Tab. 4). For the CO_2 and CH_4 source, the total annual GHG balance shifts towards a net release of 615–1990 Tg CO_2 eq. y^{-1} (result B in Tab. 4).

Table 4: Terrestrial northern high-latitude annual exchange of carbon dioxide (CO_2), methane (CH_4), and their estimated total in CO_2 equivalents (CO_2 eq.) based on a GWP_{CH4} of 25 (1 Tg CH_4 ≙ 25 Tg CO_2 eq.). Sign "–" implies a C sink and sign "+" a source of C to the atmosphere.

Kind of balance	Annual exchange in the northern high latitudes
CO_2 balance	
Case A (CO_2 sink):	–300 to –600 Tg CO_2 (eq.) y^{-1} [a]
Case B (CO_2 source):	+240 Tg CO_2 (eq.) y^{-1} [b]
CH_4 balance	
(in CO_2 eq.)	+15 to +70 Tg CH_4 y^{-1} [c] (+375 to +1750 Tg CO_2 eq. y^{-1})
Subtotals	
Result A: Case A + CH_4 balance	–225 to +1450 Tg CO_2 eq. y^{-1}
Result B: Case B + CH_4 balance	+615 to +1990 Tg CO_2 eq. y^{-1}
Total balance	**–225** to **+1990** Tg CO_2 eq. y^{-1}

[a] *AMAP* (2009); [b] *Zhuang* et al. (2006); [c] *Callaghan* et al. (2004a); *Zhuang* et al. (2004, 2006); *ACIA* (2005); *Metje* and *Frenzel* (2007); *AMAP* (2009)

Ultimately, the status of the terrestrial Arctic concerning its annual GHG balance appears to be relatively obvious: A weak net uptake of ≤ 225 Tg CO_2 eq. y^{-1} faces huge emissions of ≤ 1990 Tg CO_2 eq. y^{-1} to the atmosphere. Overall, these results suggest a positive contribution of the terrestrial northern high latitudes to global climate change at present, albeit the strength of the climatic impact varies extremely between a weak cooling (in case of the total sink) and a more or less strong warming effect (in case of the total source). The results fit into the statements of *Callaghan* et al. (2004a) and *ACIA* (2005), which come to the conclusion that respiration may outbalance assimilation on balance as CO_2- and CH_4-flux measurements reveal (*Callaghan* et al., 2004a; *ACIA*, 2005). Vast northern wetlands as for example in the W Siberian Lowlands store huge amounts of C annually due to their net CO_2 sink. Regarding both GHGs, these ecosystems act as total GHG sources in terms of CO_2 eq. owing to the CH_4 emissions (*Friborg* et al., 2003; *Callaghan* et al., 2004a; *ACIA*, 2005). At present, Arctic areas with a C source, mostly dry and mesic ecosystems (*e.g.,* E European tundra, Alaska, Isle of Svalbard), may possibly exceed those areas with a C sink, mostly wet tundra (*e.g.,* NE Greenland, N Scandinavia; *Callaghan* et al., 2004a; *ACIA*, 2005). On the other hand, some GCM simulations suggest a small C sink in the Arctic (*Callaghan* et al., 2004a; *ACIA*, 2005). The discrepancy between model results on the one hand and empirical data from field measurements on the other hand may be a result of uncertainties in upscaling local field data to regional or global scales.

6 Consequences of future climate change in the terrestrial Arctic

Global climate change is likely to have an effect on northern high-latitude ecosystems in the course of the 21st century (*e.g., McGuire* et al., 2002; *Robinson*, 2002; *IPPC*, 2007b).

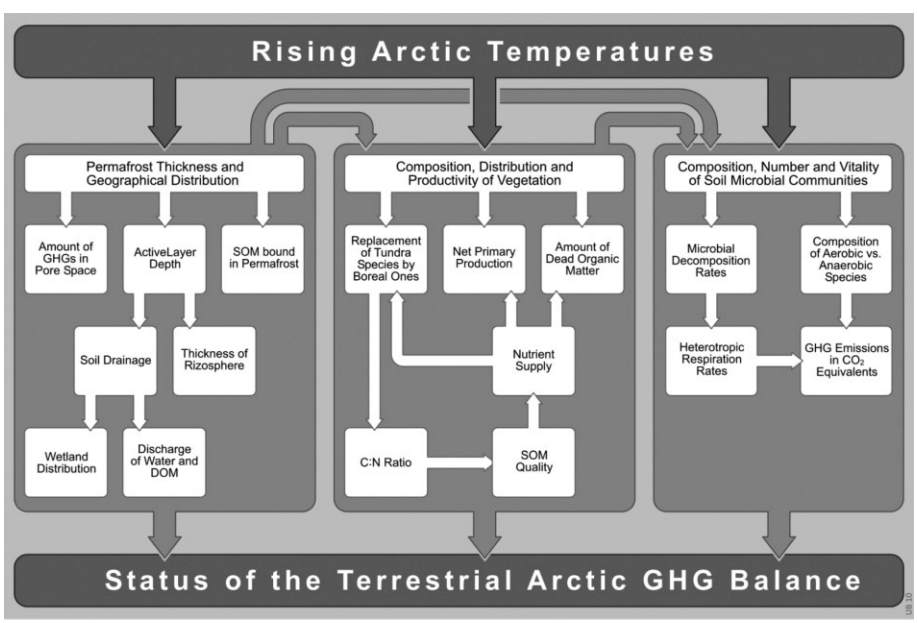

Figure 3: Simplified scheme of the impacts of rising Arctic temperatures on selected processes within the three groups of the ecosystem components permafrost, vegetation, and soil microbial communities, which determine the status of the terrestrial Arctic greenhouse-gas balance (GHG, greenhouse gases; SOM, soil organic matter; DOM, dissolved organic matter).

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Ecosystem components such as permafrost, vegetation, and soil microbes will be affected by changes in temperature and soil moisture for example. In this section, we discuss several potential impacts of climate change on (1) northern ecosystems (section 6.1) and (2) on the terrestrial Arctic C cycle and GHG exchange (section 6.2). The potential impacts on the ecosystems are discussed by highlighting the environmental variable temperature (section 6.1.1), which is a driving force for the changes being expected to occur by 2100, and the ecosystem components mentioned above (sections 6.1.2–6.1.4). Figure 3 summarizes impacts of rising Arctic temperatures on selected ecosystem processes.

6.1 Future effects on northern ecosystems

6.1.1 Consequences of changes in Arctic temperatures

Rising Arctic temperatures lead to a lengthening of the frost-free period (*Velichko* et al., 1998). As a result, longer and warmer vegetation-growing seasons (*Luo*, 2007) stimulate photosynthesis and plant growth, hence increasing vegetation's primary production (*McGuire* et al., 2002; *Rennermalm* et al., 2005; *Kimball* et al., 2006; *Stieglitz* et al., 2006; *Sitch* et al., 2007; *Luo*, 2007). On short time scales, autotrophic respiration is more sensitive to rising temperatures than photosynthetic CO_2 assimilation (*Luo*, 2007). Warming soils accelerate microbial SOM mineralization as well, at least for a short time (*Robinson* 2002; *ACIA* 2005), inducing rising respiration rates (*Hobbie* et al., 2002; *ACIA*, 2005; *Luo*, 2007). Faster decomposition rates improve nutrient supply in soils, for example the availability of nitrogen (N), which has a strong influence on the C cycle *via* plant-growth regulation (*Luo*, 2007). According to *Loya* et al. (2002), vegetation productivity and microbial activity both benefit from a growing pool of available nutrients, although in different ways (*Mack* et al., 2004). At the moment, nutrients are generally rare in many northern ecosystems owing to the unfavorable soil environment (*Hobbie* et al., 2002; *Grant* et al., 2003; *Callaghan* et al., 2004b; *ACIA*, 2005).

The frequency and intensity of wildfires and insect outbreaks are likely to increase in a warming Arctic leading to growing C emissions from combustion and decomposition (*ACIA*, 2005; *AMAP*, 2009). Wildfires liberate CO_2, thus generating a positive climate-change feedback. On the other hand, 1%–3% of the C in biomass and SOM carbon is converted during burning to black carbon (*Preston* and *Schmidt*, 2006). Since black carbon is a more refractory fraction of the SOM pool, formation of black carbon may reduce this positive feedback, and, therefore, black carbon pools have to be included in climate models (*Lehmann* et al., 2008). However, in order to project these effects for future climate change, spatially accurate information on the distribution of black carbon in soils, especially in boreal forest soils, are essential (*Lehmann* et al., 2008; *Ohlsen* et al., 2009).

6.1.2 Consequences of changes in permafrost occurrence

Permafrost development across the Arctic will be primarily affected by rising temperatures: In the 21st century, large permafrost areas may degrade in the long run, triggering a northward retreat of the remaining permafrost ultimately (*cf.* section 3). Permafrost thawing has an effect on the GHG dynamics in ecosystems *via* soil moisture and water-table position (*Callaghan* et al., 2004a; *ACIA*, 2005; *Davidson* and *Janssens*, 2006; *Rodionov* et al., 2007; *Sitch* et al., 2007). In wetlands (peatlands) and lowlands, thawing may create cooler, wetter soil conditions, while upland soils may become warmer and drier (*Callaghan* et al., 2004a; *ACIA*, 2005; *Camill*, 2005). In both cases, changing soil environment has an affect on microbial decomposition. In wetting soils, slower anaerobic mineralization prevails predominantly, favoring CH_4 production (*Davidson* and *Janssens*, 2006; *AMAP*, 2009). However, soil wetting may lead to a net C uptake *via* an intensifying photosynthetic CO_2 assimilation as experimental data have already indicated (*ACIA*, 2005). The geographical extent of northern wetlands (peatlands) as well as fluctuation of their water table have significant effects on the CH_4 being emitted (*Callaghan* et al., 2004a; *ACIA*, 2005; *Sitch* et al., 2007). Drying soils will boost CO_2 emissions (*Rodionow* et al., 2006; *AMAP*, 2009) by faster aerobic mineralization rates (*ACIA*, 2005). Altogether, drying of former permafrost soils appears to be more widespread in the Arctic in the future (*AMAP*, 2009).

Soil organic matter, which is currently bound in permafrost, may become available to decomposers in large quantities *via* thawing processes, *i.e.,* by enhancing the spatial accessibility of SOM (*Davidson* and *Janssens*, 2006; *Field* et al., 2007; *Uhlířová* et al., 2007). Another large source of potential GHGs is currently trapped in the pore space of permafrost. During the Pleistocene, ancient microbes produced GHGs that never escaped to the atmosphere being completely isolated from present biogeochemical cycles for a long time (*Steven* et al., 2006; *Rivkina* et al., 2007). The amount of CH_4 trapped in the pore space of terrestrial permafrost has been estimated to 5 Pg CH_4 by *Mitra* et al. (2005), whereas the quantities in *AMAP* (2009) range from 2 to 65 Pg CH_4. If deep-permafrost thaw occurs, these GHG might have serious implications on the Arctic GHG balance. However, their liberation takes much time as the CH_4 is located in deeper parts of the permafrost (*AMAP*, 2009).

Thawing of permafrost in combination with increasing precipitation amounts leads to more discharge of water and, hence, more DOC outflow from the terrestrial Arctic (*AMAP*, 2009). Total annual DOC-export rates from W Siberia to the Arctic Ocean for example are predicted to increase from 9.2 Tg DOC y^{-1} at present to 11.9–13.5 Tg DOC y^{-1} (*Frey* and *Smith*, 2005), which means a substantial loss from the terrestrial C reservoir. *Stieglitz* et al. (2006) simulate an increase in DOC export of 200%–400% from an Arctic Alaskan tundra ecosystem. As freshwater microbes transform aquatic DOC into C emissions of some 15–30 Tg C y^{-1} from Arctc rivers, and another 25–54 Tg C y^{-1} from lakes (*AMAP*, 2009), future increases in DOC export may lead to a growing freshwater C efflux.

6.1.3 Consequences of changes in Arctic vegetation

Climate warming is favorable to the growth, development, and reproduction of many Arctic plant species, at least transi-

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ently (*ACIA*, 2005). An increasing atmospheric CO_2 concentration stimulates photosynthesis and plant growth only for the short term (*ACIA*, 2005) as higher CO_2 levels cause a downregulation of the photosynthetic process (*Grace*, 2000, pp. 62–63; *Field* et al., 2007; *Luo*, 2007). The specific composition of the ecosystem vegetation is important for the overall magnitude of the ecosystem's response to climate warming in terms of CO_2 uptake as some plant species show no or only slightly intensified photosynthesis and growth (*Schimel*, 1995; *Luo*, 2007). Soil moisture, nutrient availability, and atmospheric CO_2 concentration have an influence on Arctic plants' development, too (*ACIA*, 2005). Warming-induced changes in nutrient supply are a potential key to an increasing vegetation productivity (*Shaver* et al., 2000; *Hobbie* et al., 2002; *Robinson*, 2002; *Callaghan* et al., 2004b; *Marchand* et al., 2004; *ACIA*, 2005). Increasing turnover of C and nutrients such as N owing to soil warming may transfer many essential elements from the soil pool into the vegetation pool (*Uhlířová* et al., 2007). Increasing C sequestration by plants may alter the C : N ratio of their tissue in turn (*ACIA*, 2005; *Luo*, 2007), affecting SOM quality and decomposition rates (*Robinson*, 2002; *Cornelissen* et al., 2007). Slower mineralization rates of woody boreal plant species compared to tundra species (*Steltzer*, 2004) may reduce overall ecosystem C losses (*Cornelissen* et al., 2007; *Luo*, 2007). The coinstantaneous decline in nutrient release from SOM, *e.g.,* N, is likely to limit the growth of woody species in turn (*Chapin* et al., 2005; *Cornelissen* et al., 2007). Accelerated plant growth as well as changes to woody species may limit the availability of N in the long run (*Luo*, 2007), ultimately restricting primary production and, thus, C sequestration over time.

Timing, depth, and duration of winter snow cover is one of the fastest changing climate features (*IPCC*, 2007a). Snow cover controls plant-growing conditions, *i.e.,* phenology, productivity, and community composition, mainly by defining the start and duration of the potential growing season (*e.g., Odland* and *Munkejord*, 2008). The Arctic species group with the greatest phenological response to snowsmelt changes are dwarf shrubs as revealed in long-term snow-fence experiments (*Wipf* and *Rixen*, 2010). Additionally, their abundance also increases.

The current distribution of many Arctic plants may change in the course of this century due to the northward expansion of boreal species in response to climate warming (*Kaplan* et al., 2003; *ACIA*, 2005; *Cornelissen* et al., 2007; *Luo*, 2007). A northward migration of boreal shrubs and trees into many southern tundra ecosystems has already been observed in the last decades (*Sturm* et al., 2001; *Callaghan* et al., 2004a; *ACIA*, 2005; *Cornelissen* et al., 2007; *AMAP*, 2009). In combination with a more intense growth of existing tundra plants, these changes have already increased the total vegetation productivity in the Arctic (*Sitch* et al., 2007). The increasing number and vitality of the Arctic plants led to the recent "greening of the Arctic", which has been measured by a higher normalized difference vegetation index (NDVI; *Callaghan* et al., 2004a; *ACIA*, 2005; *Hinzman* et al., 2005; *Richter-Menge* et al., 2006; *Cornelissen* et al., 2007). However, an overall greening trend for the entire Arctic area is not consistent: While the NDVI of tundra ecosystems in N America increased by ≈ 10% on average over the last two decades, boreal forest NDVI declined owing to effects of drought (*Richter-Menge* et al., 2006; *Sitch* et al., 2007; *AMAP*, 2009). Additionally to the greening of the Arctic, the northward expansion of the biologically more productive woody boreal vegetation is likely to increase the primary production in this region, filling vegetation C stocks (*AMAP*, 2009).

A geographical displacement of ecosystems, and potentially even of entire biomes, in this context may be possible (*Kaplan* et al., 2003). At least an advance of the polar tree line is predicted to occur in several parts of the Arctic (*Callaghan* et al., 2004a; *ACIA*, 2005; *Sitch* et al., 2007), or even throughout the entire Arctic area (*Kaplan* et al., 2003). The corresponding extension of the taiga towards north is simulated by the biogeochemistry-biogeography model "BIOME4" (*Kaplan* et al., 2003). Recent results of several global vegetation models and regional models also indicate a displacement of the current tundra area by the advancing taiga of ≈ 11%–50% in case of a double CO_2 concentration ($2 \times CO_2$) scenario (*Callaghan* et al., 2004a; *ACIA* 2005), or taking the IS92a ("business-as-usual") emission scenario as a basis (*White* et al., 2000). The "Lund-Potsdam-Jena Dynamic Gobal Vegetation Model" (LPJ-DGVM) projects an analogous extension of the boreal area of ≈ 11.3% (total range: 9.8%–14.4%) for the period 1960–2080 (*Sitch* et al., 2003). In the High Arctic, ≈ 17.6% (total range: 14.2%–23.0%) of the current polar desert area may become replaced by tundra ecosystems during that time (*Sitch* et al., 2003). Due to high migration rates of local habitats, the loss of local species in Arctic and mountain areas might be 6%–20% regarding the $2 \times CO_2$ scenario (*Malcolm* and *Markham*, 2000). Such changes might be realized within 200–300 y (*Kaplan* et al., 2003).

6.1.4 Consequences of changes in Arctic soil microbes

A warming Arctic climate and its impacts on permafrost and vegetation may have major effects on the activity of microbes in northern soils. As the prerequisites for life improve, microbial communities are expected to evolve under the new conditions, finally growing in population (*Vorobyova* et al., 1997). The microbial "revitalization" is a result of the restoration of a more active metabolism in warming soils (*Vorobyova* et al., 1997). Ameliorating life conditions do not only stimulate the activity of existing microbial species, they also induce alterations within the communities (*ACIA*, 2005; *Luo*, 2007). Soil-organic-matter mineralization and C respiration will both benefit from increasing microbial activity, likely releasing more C to the atmosphere (*Hobbie* et al., 2002; *Mikan* et al., 2002; *Marchand* et al., 2004; *ACIA*, 2005); after a while, respiration rates decrease as microbes acclimate to higher temperatures and/or microbial substrate supply diminishes (*Luo*, 2007). Isolated permafrost communities will slowly be able to participate again in current biogeochemical cycles including the production, consumption and release of GHGs (*Khmelenina* et al., 2002; *Trotsenko* and *Khmelenina*, 2005; *Rivkina* et al., 2007). Regarding an increasing soil temperature exclusively, and assuming all other environmental factors to level off, aerobic and anaerobic microbial respiration rates are likely to increase (*Mikan* et al., 2002; *Rivkina* et al., 2002, 2007; *Metje*

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and *Frenzel*, 2007; *Rodionov* et al., 2007). Thus, CH_4 production by methanogens may exhibit substantial increases (*Wagner* et al., 2007), leading to larger CH_4 emissions from northern wetlands if Arctic temperatures continue to rise and all other environmental factors remain constant (*Topp* and *Pattey*, 1997; *Blodau*, 2002; *Callaghan* et al., 2004b; *ACIA*, 2005). At the same time, CH_4 oxidation to CO_2 by methanotrophs may increase as well under aerobic conditions (*Khmelenina* et al., 2002; *Rivkina* et al., 2007). In fact, methanogensis is much more sensitive to temperature increase than methanotrophy (*Topp* and *Pattey*, 1997; *Blodau*, 2002; *Callaghan* et al., 2004b; *ACIA*, 2005; *Whalen*, 2005) and aerobic respiration (*Blodau*, 2002). According to that, CH_4 production may increase relative to CH_4 oxidation and CO_2 respiration. However, soil moisture and water-table position have an essential influence on actual rates of these microbial processes. For that reason, the specific conditions of soil temperature and soil moisture (water-table position) will finally determine the magnitude of the exchange of both gas species by Arctic ecosystems.

6.2 Future effects on the total greenhouse-gas balance

Regarding both GHGs, the northern high latitudes are presumably a present annual source of C to the atmosphere as our calculations have indicated. Data on C fluxes support this view, although some GCMs project a sink (*cf.* section 5.2). Whether northern ecosystems remain a source of C to the atmosphere or whether they become a C sink in the future depends on the ratio of C assimilation to C release, and how they will respond to climate warming relative to one another (*ACIA*, 2005; *Sitch* et al., 2007).

If conditions improve for plant CO_2 assimilation primarly, C sequestration may equal or may even outbalance respiratory C losses (positive NEP), producing a negative feedback to climate warming (*Hobbie* et al., 2002; *Luo*, 2007). Present C losses from high-latitude ecosystems may turn into a weak C sequestration, as indicated by *Callaghan* et al. (2004a) and *ACIA* (2005). Prognostic model simulations for different projected temperature increases resulted in the lowest C gain in case of the warmest emission scenario, while the coldest scenario led to the highest C storage (*Callaghan* et al., 2004a; *ACIA*, 2005; *Sitch* et al., 2007). Using several models, a C gain has been simulated for the Arctic in the 21st century (see Tab. 5).

On the other hand, some studies suggest future C losses from the Arctic. If climate warming is more favorable to decomposition and respiration, C will be lost from Arctic ecosystems (*Luo*, 2007). *Mack* et al. (2004) assumed that rising emissions may add to the greenhouse effect due to the fact that soil warming does not necessarily initiate an increasing net C gain. In their fertilization experiments, plant productivity doubled indeed, though an increasing bioavailability of nutrients favored microbial respiration more than primary production. Overall, a substantial release of C (and nutrients such as N) from soil has been observed, exceeding photosynthetic C uptake (*Mack* et al., 2004). The net C loss from this Alaskan tundra ecosystem where *Mack* et al. (2004) conducted their study reached almost 2000 g C m^{-2} within 20 y. Simulations of soil warming by Panikov (1994, 1998 in *ACIA,* 2005) suggest that decomposition may be more affected by rising temperatures, lowering soil C pools; this corresponds to the findings of *Mack* et al. (2004). *Stieglitz* et al. (2006) use a simplified "plant–soil model" (PSM), which projects an increase in C sequestration by plants (200–300 g C m^{-2}) and C loss from soils (200–600 g C m^{-2}) in Arctic Alaska during the time period 2000–2100. Consequently, a net C loss from this tundra ecosystem has been simulated.

Prognostic simulations by *Zhuang* et al. (2006) for the high latitudes above 50° N and the period 1991–2100 using the TEM, yield a net C efflux (based on both GHGs), which increases from 276 Tg C y^{-1} at present to 473 Tg C y^{-1} by the end of this century. In these simulations, the cumulative net C efflux between 1991 and 2100 ranges from 34.38 to 54.12 Pg C under scenarios of low to high anthropogenic emissions, and without any fertilization effects of a rising atmospheric

Table 5: Carbon (C) gain in the terrestrial Arctic in the 21st century as projected by several studies using circulation models.

Study	Circulation model	Geographical region	Prognostic time period	Projected carbon gain
McGuire et al. (2000)	TEM[a]	50°–90° N	1995–2100	mean: 16 g C m^{-2} y^{-1} range: –1 to +32 g C m^{-2} y^{-1}
Clein et al. (2000)	TEM[a] GEM[b]	50°–90° N 50°–90° N	1995–2100 1995–2100	cumulative: 2032 g C m^{-2} cumulative: 463 g C m^{-2}
Euskirchen et al. (2006)	TEM[a]	60°–90° N	2001–2100	increase: from 0.07 Pg C y^{-1} (2000s) to 0.46 Pg C y^{-1} (2090s)
White et al. (2000)	hybrid v4.1[c]	50°–90° N	2000–2100	cumulative 1860–2100: 134 Pg C 2000–2050: 0.4 Pg C y^{-1} 2050–2100: 0.8–1.0 Pg C y^{-1}
Sitch et al. (2003)	LPJ-DGVM[d]	60°–90° N	1960–2080	cumulative mean: 18.3 Pg C cumulative range: 12.2–31.3 Pg C

[a] Terrestrial Ecosystem Model; [b] General Ecosystem Model; [c] Gobal Vegetation Model; [d] Lund-Potsdam-Jena Dynamic Global Vegetation Model

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CO_2 concentration. With CO_2-fertilization effects, the cumulative net source is 23.37–33.43 Pg C for low anthropogenic emissions, 4.60–22.10 Pg C for intermediate emissions, and finally turns into a net C uptake of 4.86–24.13 Pg C regarding high emissions (*Zhuang* et al., 2006). *Gruber* et al. (2004, p. 59f.) estimate the amount of potential C release from permafrost and global wetlands by 2100 to 100 Pg C in each case. According to *Grant* et al. (2003) and *Zhuang* et al. (2004), climate warming is likely to increase present CH_4 emissions from Arctic ecosystems (17–70 Tg CH_4 y^{-1}) in the future. As most CH_4 originates from northern wetlands (peatlands), this prognosis is largely consistent with the global trend for wetland emissions: On the global scale, present wetland CH_4 efflux (60–237 Tg CH_4 y^{-1}) is predicted to rise to 500–600 Tg CH_4 y^{-1} by the end of this century using a simple CH_4 emission scheme (*Gedney* et al., 2004). If climate warming is more favorable to CH_4 release from wetlands compared to CH_4 consumption in upland soils (*Sitch* et al., 2007), the total CH_4 emission from the terrestrial Arctic will increase overall. In terms of CO_2 eq., they may generate a net GHG source as a result of the high GWP of CH_4 (*Sitch* et al., 2007).

Beyond the C emissions, positive radiative forcing impacts of increasing water vapour and wildfires as well as albedo-temperature feedbacks of shifting biomes will likely offset (*Field* et al., 2007) or even dominate (*McGuire* et al., 2006) the effects of increasing vegetation C storage. Anthropogenic changes in land use are another mechanism of positive radiative forcing to add to climate warming (*Field* et al., 2007). A complete replacement of tundra ecosystems by the advancing boreal forests might bring a C gain of 21 Pg C, whereas the albedo effect of the conversion over the same area might be equivalent to an additional emission of 32 Pg C (*Field* et al., 2007). Moreover, forest migration might be connected with emissions of 100 Pg C from thawing permafrost, and 2–10 Pg C as CH_4 (*Field* et al., 2007). Hence, the continual expansion of the taiga would be responsible for a C source in the northern high latitudes that would contribute to climate warming (*Field* et al., 2007).

In general, northern ecosystems are likely to respond in different ways and to varying degrees to the upcoming climatic changes owing to regional differences (*Christensen* et al., 2004). The geographical extent of northern wetlands (peatlands) and the position of their water tables are further critical factors in terms of the total GHG balance due to their potential to emit CH_4 in huge quantities (*Callaghan* et al., 2004b; *ACIA*, 2005; *Sitch* et al., 2007).

7 Concluding remarks regarding greenhouse-gas balances

Current C fluxes will assimilate to changing environmental conditions in the future, and ecosystem C uptake will rise or decline in relation to ecosystem C respiration. Present total (CO_2 and CH_4) flux data from the Arctic vary between a weak storage of ≤ 225 Tg CO_2 eq. y^{-1} and a huge release of ≤ 1990 Tg CO_2 eq. y^{-1} regarding a GWP_{CH4} of 25. These results, which suggest a positive contribution of the terrestrial Arctic to global climate change, correspond to the statement of *ACIA* (2005) about a present source of C and radiative forcing in the Arctic. Estimates of future CO_2 and CH_4 exchange vary widely. Several studies predict an Arctic that may become a C sink (*e.g., McGuire* et al., 2000; *White* et al., 2000; *Sitch* et al., 2003; *ACIA*, 2005; *Euskirchen* et al., 2006), counteracting climate change. On the other hand, several studies expect a net loss enhancing current climatic changes (*e.g., Gruber* et al., 2004, p. 59f.; *Zhuang* et al., 2006; *Field* et al., 2007; *Sitch* et al., 2007).

The controversial views about the future status of the Arctic in terms of its annual GHG balance are the result of (1) different measurement techniques (*e.g.,* eddy covariance *vs.* chamber technique), whose results are incorporated into models that are (2) based on varying scenarios of climate change (*e.g.,* A1B, A2, B2, IS92a, 2 × CO_2) and/or (3) compute projections for different time periods and/or (4) geographical regions. And finally, the controversy reflects the high complexity and variability of the C fluxes in northern high-latitude ecosystems over time and space. In order to improve our knowledge about the present and future status of the terrestrial Arctic C cycle including the total GHG balance, we need more field-observation data on C dynamics from multiple spatial and temporal scales, particularly from underrepresented locations, as well as multiscale experiments.

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